

The Intermission Challenge Group 7

ETF3480 - ETF5480 Assignment 2

Tzu-Hsuan Yang (35461586)

Nutchanok Saitin (35493011)

Mohammad Zulkifli Falaqi

Bagas Wibawanto

Table of contents

1. Introduction and background	2
2. Analytical Methods	2
3. Results	2
3.1 Model Parameter and Assumptions	2
3.1.1 Input Parameters	2
3.1.2 Modelling Assumptions	3
3.2 Baseline System Performance	4
3.2.1 Configuration 1: Status Quo	4
3.2.2 Configuration 2: All Unisex	5
3.2.3 Configuration 3: Hybrid Model	5
3.2.4 Comparative Equity Analysis	6
3.3 System Behaviour Under Surge Demand	7
3.3.1 Early Surge with Male-Dominated Arrivals	7
3.3.2 Early Surge with Female-Dominated Arrivals	7
3.3.3 Late Surge with Male-Dominated Arrivals	7
3.3.4 Late Surge with Female-Dominated Arrivals	7
3.4 Comparative Equity Analysis Under Surge Demand	7
3.4.1 Operational performance across surge scenarios	7
3.4.2 Fairness and equity scores under various surge scenarios	7
3.4.3 Overall ranking and robustness analysis	7
4. Discussions of results and business recommendation	8
5. Conclusion and further suggested work	8

1. Introduction and background

During performances, the concert hall has a strict 25-minute intermission. This short period can create a sudden rush to restroom facilities and cause serious congestion. Management has received increasing complaints about long waiting times and the lack of inclusive restroom design. Therefore, this problem is not only about reducing queues, but also about improving fairness and accessibility for different users.

Historical observations show that restroom users do not all behave in the same way. Users can have different service-time needs and different preferences for restroom types. Some users prefer gender-segregated facilities, some have no strong preference, and some prefer unisex facilities. Because of this, the best restroom design should not only have shorter average waiting times, but should also consider whether different user groups are treated fairly.

This report evaluates three possible restroom configurations: the Status Quo, the All Unisex model, and the Hybrid model. The analysis uses queueing theory to estimate waiting-time performance, simulation to test more realistic intermission conditions, and fairness measures to compare the experience of different user groups.

The three configurations evaluated in this report are:

1. **Status Quo**: 20 women's stalls, 10 men's urinals, and 10 men's stalls.
2. **All Unisex**: 40 universal all-gender stalls.
3. **Hybrid Model**: 15 women's stalls, 10 men's urinals, 5 men's stalls, and 10 unisex stalls.

The aim of this report is to provide management with a clear recommendation about which configuration should be used. The recommended design should reduce congestion during intermission while also supporting a more inclusive restroom system.

2. Analytical Methods

3. Results

The analysis proceeds in two phases. First, a closed-form analytical model establishes baseline performance metrics and equity scores for each configuration under idealised steady-state conditions. These results are then stress-tested through discrete-event simulation to examine system behaviour under realistic intermission demand patterns.

3.1 Model Parameter and Assumptions

Before presenting results, this section summarises the input parameters and modelling assumptions applied consistently across all configurations and scenarios. These values are derived directly from the historical observation data provided in the case.

3.1.1 Input Parameters

Table 1 — Observed Input Parameters

Parameter		Women	Men
Arrival rate (Poi, pers/min)		3.2	3.2
Inter-arrival time (Exp, min)		0.3125	0.3125
Probability (Class 1)		95%	95%
Class 1 Service Time (min)	Mean	1.67	0.83
	SD	1.2	0.7
Probability (Class 2)		5%	5%
Class 2 Service Time (min)	Mean	5.2	5.2
	SD	3.2	3.2

Table 2 — User Preference Distribution

Preference Type	Proportion
Gender-segregated (GS)	70%
No preference (NP)	21%
Unisex (UX)	9%

Applies equally to both genders and across all configurations.

Table 3 — User Satisfaction Score Table

Wait Bracket (min)	Gender:	Women						Men					
	Preference:	GS		No-Pref		Unisex		GS		No-Pref		Unisex	
	Restroom:	GS	Unisex	GS	Unisex	GS	Unisex	GS	Unisex	GS	Unisex	GS	Unisex
$0 \leq t < 5$		5	5	3	4	5	5	5	5	3	4	5	5
$5 \leq t < 10$		4	4	2	3	4	4	4	4	2	3	4	4
$10 \leq t < 20$		3	2	1	2	2	3	3	2	1	2	2	3
$20 \leq t < 40$		1	1	0	1	1	2	2	1	0	0	1	2
$t \geq 40$		0	0	0	0	0	1	0	0	0	0	0	1

Table 4 — Configuration Fixture Allocation

Restroom Type	Config 1	Config 2	Config 3
Women's stalls	20	-	15
Men's urinals	10	-	10
Men's stalls	10	-	5
Unisex stalls	-	40	10
Total	40	40	40

3.1.2 Modelling Assumptions

The following assumptions are applied throughout the analysis:

- Arrivals follow a Poisson process. The concert hall has a fixed total audience capacity, of which a certain proportion are expected to seek restroom access during the intermission, collectively reflected in an aggregate arrival rate of 6.4 pers/min across all genders. Each gender's individual arrival rate is then derived proportionally from this aggregate based on the assumed gender composition of the audience.
- Service times are treated as a general distribution, characterised by the empirically observed mean and SD from the historical data.
- For the baseline analysis, no-preference users are assumed to split equally (50-50) between gender-segregated and unisex facilities where available. In the simulation scenarios, this routing rule is replaced by a shortest queue length rule or shortest expected wait time rule, reflecting more realistic decision-making behaviour under congestion.
- Gender-diverse patrons are acknowledged qualitatively but excluded from quantitative scoring due to the absence of arrival rate data in the historical observations.
- Steady-state conditions are assumed for the baseline analytical model.

3.2 Baseline System Performance

Before presenting results, this section summarises the input parameters and modelling assumptions applied consistently across all configurations and scenarios. These values are derived directly from the historical observation data provided in the case.

3.2.1 Configuration 1: Status Quo

Queuing performance

Line	k	λ (/min)	μ (/min)	CV	ρ (%)	Wq (min)	Lq	W (min)	L
Women's stalls	20	3.200	0.5418	0.8515	29.54	0.0000890	0.000285	1.8466	5.909
Men's urinals	10	3.040	1.2048	0.8434	25.23	0.000386	0.001173	0.8304	2.524
Men's stalls	10	0.160	0.1923	0.6154	8.32	0.0000209	~0.000003	5.2000	0.832

Fairness scores

Gender	Preference	Facility Used	Wq (min)	Bracket	Score
Women	GS pref (70%)	GS	0.0000890	0	5
Women	No pref (21%)	GS	0.0000890	0	5
Women	UX pref (9%)	GS (forced)	0.0000890	0	3
Men — urinal	GS pref (70%)	GS	0.000386	0	5
Men — urinal	No pref (21%)	GS	0.000386	0	5
Men — urinal	UX pref (9%)	GS (forced)	0.000386	0	3
Men — stall	GS pref (70%)	GS	0.0000209	0	5
Men — stall	No pref (21%)	GS	0.0000209	0	5
Men — stall	UX pref (9%)	GS (forced)	0.0000209	0	3

Metrics	Score
Totalitarian (Tr)	4.82
Rawlsian (Mr)	3

3.2.2 Configuration 2: All Unisex

Queuing performance

Line	k	λ (/min)	μ (/min)	CV	ρ (%)	Wq (min)	Lq	W (min)	L
Unisex stalls	40	6.400	0.6908	1.0562	23.16	0.00000872	0.0001	1.4475	9.264

Fairness scores

Gender	Preference	Facility Used	Wq (min)	Bracket	Score
Women	GS pref (70%)	UX (forced)	0.00000872	0	4
Women	No pref (21%)	UX	0.00000872	0	5
Women	UX pref (9%)	UX (matched)	0.00000872	0	5
Men	GS pref (70%)	UX (forced)	0.00000872	0	4
Men	No pref (21%)	UX	0.00000872	0	5
Men	UX pref (9%)	UX (matched)	0.00000872	0	5

Metrics	Score
Totalitarian (Tr)	4.3
Rawlsian (Mr)	4

3.2.3 Configuration 3: Hybrid Model

Queuing performance

Line	k	λ (/min)	μ (/min)	CV	ρ (%)	Wq (min)	Lq	W (min)	L
Women's stalls	15	2.576	0.5418	0.8515	31.70	0.000600	0.001546	1.8471	4.758
Men's urinals	10	2.447	1.2048	0.8434	20.31	0.000175	0.000428	0.8302	2.032
Men's stalls	5	0.129	0.1923	0.6154	13.40	0.004440	0.000573	5.2044	0.671
Unisex stalls	10	1.248	0.6908	1.0562	18.06	0.000254	0.000317	1.4478	1.807

Fairness scores

Gender	Preference	Facility Used	Wq (min)	Bracke t	Score
Women	GS pref (70%)	GS	0.000696	0	5
Women	No pref -> GS (10.5%)	GS	0.000696	0	5
Women	No pref -> UX (10.5%)	UX	0.000240	0	5
Women	UX pref (9%)	UX (matched)	0.000240	0	5
Men	GS pref (70%)	GS urinal	0.000204	0	5
Men	No pref -> GS (10.5%)	GS urinal	0.000204	0	5
Men	No pref -> UX (10.5%)	UX	0.000240	0	5
Men	UX pref (9%)	UX (matched)	0.000240	0	5

Metrics	Score
Totalitarian (Tr)	5
Rawlsian (Mr)	5

3.2.4 Comparative Equity Analysis

In general, all configurations perform well in terms of overall average waiting time, with all lines operating well within capacity. It is expected since this is a baseline analysis under idealised steady-state conditions, where the system is not stressed by burst arrivals or high service-time variability. The M/G/k model is applied to all configurations using the pooled WAST parameters. Results show all systems are well within capacity ($\rho < 35\%$ across all lines), with near-zero wait times.

However, the fairness scores reveal important differences in how each configuration serves different user groups. Across the three configurations, the equity scores can be summarised as follows:

Metric	Config 1	Config 2	Config 3
Totalitarian (Tr)	4.82	4.30	5
Rawlsian (Mr)	3	4	5

Under these conditions, the fairness scores are driven entirely by access and preference match rather than wait times. Config 3 achieves perfect Tr and Mr. Config 2 with Unisex only restrooms has a lower Tr due to the lower satisfaction scores for gender-segregated preference users (majority), but a higher Mr than Config 1 because it provides better access for unisex-preference users. Config 1 has a higher Tr than Config 2 because it provides better access for the majority gender-segregated preference users, but a lower Mr than Config 2 because it provides no access for unisex-preference users.

Based on the resulted equity scores, the configurations can be ranked as follows:

- Totalitarian ranking: Config 3 > Config 1 > Config 2
- Rawlsian ranking: Config 3 > Config 2 > Config 1

While the baseline results establish a clear ranking under idealised conditions, real concert hall intermissions rarely conform to steady-state assumptions. Patrons tend to surge toward restrooms immediately after the performance pauses, or rush back just before it resumes. The following section examines whether the baseline ranking holds under these more realistic demand patterns.

3.3 System Behaviour Under Surge Demand

Four surge scenarios are simulated to stress-test the baseline findings. All scenarios apply a burst arrival pattern where 80% of patrons arrive within a concentrated 5-minute window, with the remaining 20% spread across the rest of the intermission. Two timing variants are tested, early burst (first 5 minutes) and late burst (last 5 minutes), each combined with two gender compositions: male-dominated and female-dominated arrivals.

The gender composition scenarios reflect a practical reality of concert hall operations: audience demographics are not always balanced and are often shaped by the nature of the performance. A K-pop or idol group concert, for instance, typically attracts a predominantly female audience, while other genres may skew male. Rather than assuming a 50-50 gender split, these scenarios examine how each configuration performs when one gender accounts for 80% of total arrivals, capturing the service time variance and diverse stakeholder needs that arise from such demographic shifts.

This stress test approach uses the pareto principle to create a more extreme scenario, where the majority of demand is concentrated in a short time window, and one gender dominates the arrivals. This allows us to evaluate the robustness of each configuration under conditions that are more challenging than the baseline steady-state scenario, and to see how well they can accommodate different user groups when demand is highly concentrated.

	Early Burst (first 5 min)	Late Burst (last 5 min)
Male-Dominant (80% Men)	Scenario 1	Scenario 3
Female-Dominant (80% Women)	Scenario 2	Scenario 4

3.3.1 Early Surge with Male-Dominated Arrivals

3.3.2 Early Surge with Female-Dominated Arrivals

3.3.3 Late Surge with Male-Dominated Arrivals

3.3.4 Late Surge with Female-Dominated Arrivals

3.4 Comparative Equity Analysis Under Surge Demand

3.4.1 Operational performance across surge scenarios

3.4.2 Fairness and equity scores under various surge scenarios

3.4.3 Overall ranking and robustness analysis

Average waiting time by configuration: bar chart

Women vs men waiting time: gender disparity

Worst-group waiting time by configuration: Direction A□

Equity score comparison: preference table into equity scores

Burst arrival results: table or line chart eg. queue length over time or users still waiting after 25 minutes

Apply a fairness feasibility threshold: a configuration is considered unacceptable if any stakeholder group has an average waiting time greater than 15 minutes, because this represents more than half of the 25-minute intermission and would leave insufficient time for patrons to return comfortably before the performance resumes.

- Primary threshold: any group average wait > 15 minutes is unacceptable.
- Secondary stress indicator: if many users remain unserved after 25 minutes, the configuration is not robust.

threshold can be adjusted, now just for example

4. Discussions of results and business recommendation

5. Conclusion and further suggested work

NOT FINALISED

1. Introduction and Background

1.1 Business problem: 25-minute intermission and restroom congestion

During performances, the concert hall has a strict 25-minute intermission. This short period creates a sudden rush to restroom facilities, leading to congestion and long waiting times. Management has received increasing complaints not only about queue length, but also about the lack of inclusive restroom design.

1.2 Why this is also a fairness and inclusivity problem

Different users have different needs and preferences. Some prefer gender-segregated facilities, some have no strong preference, and some prefer unisex facilities. Because of this, the best restroom design should not only have shorter average waiting times, but should also consider whether different user groups are treated fairly.

1.3 Report scope: Primary Direction A + secondary Direction B

This report focuses primarily on **Direction A: Fairness and Stakeholder Disparity**. The three baseline configurations are compared using both waiting-time performance (Totalitarian) and preference-based equity measures (Rawlsian). As a secondary sensitivity analysis, the report also considers **Direction B: Splitting Rules**, by examining how different assumptions about no-preference users may affect the results.

1.4 Three configurations compared

The three configurations evaluated in this report are:

1. **Status Quo**: 20 women's stalls, 10 men's urinals, and 10 men's stalls.
2. **All Unisex**: 40 universal all-gender stalls.
3. **Hybrid Model**: 15 women's stalls, 10 men's urinals, 5 men's stalls, and 10 unisex stalls.

(If we decide to include Direction C, we can add it here and in the analysis sections below.)

2. Analytical Methods

2.1 Queueing model assumptions

- Poisson arrivals
- Non-Exponential / variable service times
- Multi-server queues

2.2 Input parameters and stakeholder groups

Stakeholder	Meaning
Women, gender-segregated preference	Prefer women's facilities
Men, gender-segregated preference	Prefer men's facilities
No-preference users	Can use either gender-segregated or unisex
Unisex-preference users	Prefer unisex facilities
Class 1 users	Shorter service time
Class 2 users	Longer service time

2.3 Direction A fairness framework

- Average wait: Efficiency
- Group wait: Stakeholder disparity
- Worst-group wait: Rawlsian fairness
- Equity score: Inclusive design / user satisfaction
- Difference between longest and shortest group wait: Inequality / disparity
- Totalitarian vs Rawlsian approach
 - Totalitarian criterion: Which configuration gives the highest total welfare or lowest overall wait
 - Rawlsian criterion: Which configuration minimises the worst stakeholder waiting time or maximises the minimum stakeholder welfare

2.4 Direction B splitting rule sensitivity

- Focus on how no-preference users are split between gender-segregated and unisex facilities,
- Baseline fixed split: No-preference users are assumed to be randomly split 50/50 between gender-segregated and unisex facilities.
 - More GS-oriented split: No-preference users mostly follow gender-segregated facilities
 - More unisex-oriented split: No-preference users more willing to use unisex facilities
 - Shortest queue rule: No-preference users choose the shorter queue at the time of arrival
 - Shortest expected wait rule: No-preference users choose the option with lower expected wait

Assumption: We only apply adaptive splitting to no-preference users, while gender-segregated preference and unisex-preference users mainly follow their stated preferences.

2.5 Simulation scenarios

The simulation is used to test whether the analytical queueing results remain reliable under more realistic intermission conditions. In particular, the simulation considers three scenarios: - Baseline: Based on the historical arrival and service-time parameters provided in the case, the simulation assumes that arrivals follow the baseline stochastic process and that users follow the initial splitting rule. (As the normal intermission scenario to think which configuration performs best under typical conditions.) - Burst arrival: 80% arrive in first 5 minutes (To stress-test the restroom system under a realistic intermission rush scenario, where many users arrive at the same time. Which configuration is most robust when demand is highly concentrated at the beginning of the intermission?) - Higher service-time variance: The baseline service-time distribution may underestimate the diversity of user needs. Some users may require more time due to accessibility needs, caregiving responsibilities,

clothing, mobility limitations, or other personal factors. Therefore, this scenario increases the variability of service times while keeping the same basic arrival structure. (To test whether the recommended configuration remains fair when service times are more unequal. Which configuration is less sensitive to service-time variability?) <- Important for fairness because a system that only works well when most users have short service times may not be inclusive.

Metric:

mean wait - Average waiting time across users: overall efficiency

median wait - Typical user's waiting time: less sensitive to outliers than mean

90th percentile wait - Waiting time experienced by the worst 10% of users: measures fairness a

maximum wait - Longest waiting time experienced by any user: extreme congestion risk

percentage of users served within 25 minutes - percentage of users who complete restroom u

number of users still waiting at end of intermission - Number of users not served before perfo

3.2 Analytical waiting-time results

3.3 Simulation results

3.4 Fairness and equity-score results

3.5 Burst-arrival stress test

3.6 Direction B sensitivity results

Average waiting time by configuration: bar chart

Women vs men waiting time: gender disparity

Worst-group waiting time by configuration: Direction A

Equity score comparison: preference table into equity scores

Burst arrival results: table or line chart eg. queue length over time or users still waiting a

Apply a fairness feasibility threshold: a configuration is considered unacceptable if any stakeholder group experiences a substantially longer waiting time during a 15-minute intermission and would leave insufficient time for patrons to return comfortably before the next service period.

- Primary threshold: any group average wait > 15 minutes is unacceptable.
- Secondary stress indicator: if many users remain unserved after 25 minutes, the configuration is unacceptable.

threshold can be adjusted, now just for example

4. Discussion and Business Recommendation (Example)

4.1 Efficiency comparison

In terms of overall average waiting time, Configuration X performs best, followed by Configuration Y and Z.

4.2 Fairness comparison

Although a configuration may be efficient on average, it may still be unacceptable if one stakeholder group experiences a substantially longer waiting time.

4.3 Apply decision threshold - Reject configurations where any group average wait > 15 minutes

Criterion	Meaning
Efficiency	Overall average wait should be low
Fairness	Worst-group wait should not exceed threshold
Inclusivity	Unisex-preference users should have reasonable access
Robustness	Performance should not collapse under burst arrival
Practicality	Configuration should be operationally feasible

If a configuration has a lowest avg waiting time, but one group has an average wait of 15 minutes, it would be rejected based on the fairness criterion. If a configuration considered as fair but collapses under burst arrival, it would be rejected based on the robustness criterion. If a configuration is efficient and fair, but has no unisex access, it would be rejected based on the inclusivity criterion. The final recommendation should be the configuration that best balances these criteria.

4.4 Discuss inclusivity and unisex access

Unisex facilities provide flexibility and support inclusive access, especially for users who prefer or require all-gender facilities. However, if unisex capacity is too small, these users may experience long queues and low equity scores. Therefore, the recommendation should consider both the number of unisex units and how users are directed to them during peak demand.

The conclusion is also tested under alternative splitting rules. If no-preference users are willing to move toward the shorter queue, the Hybrid Model may perform better because unisex stalls can absorb overflow demand. However, this depends on clear signage and user awareness.

4.5 Recommend final configuration

4.6 Explain why other options are not selected

What does it mean for management, and what should they do?

5. Conclusion and Further Suggested Work: Summarise final recommendation and acknowledge limitations.

5.1 Final summary of recommendation

Based on the combined evaluation of waiting-time performance, stakeholder disparity, equity scores, and burst-arrival robustness, this report recommends **Configuration X** as the preferred design. The optimal configuration does not mean it is perfect in every aspect, but it avoids having any stakeholder group experiencing an average wait time greater than 15 minutes, while also providing reasonable access for unisex-preference users and maintaining robustness under burst arrival conditions.

5.2 Main limitations

5.3 Further work - More real data: Improve accuracy of arrival assumptions - Cost analysis: Assess feasibility of redesign - Different event types: Audience behaviour may differ by concert type - More behavioural split rules: Better represent real user decisions - Accessibility and sustainability considerations: Consider how design affects users with disabilities and environmental impact of different configurations

6. Appendix

- Code
- Full simulation details
- Extra tables/figures

- References